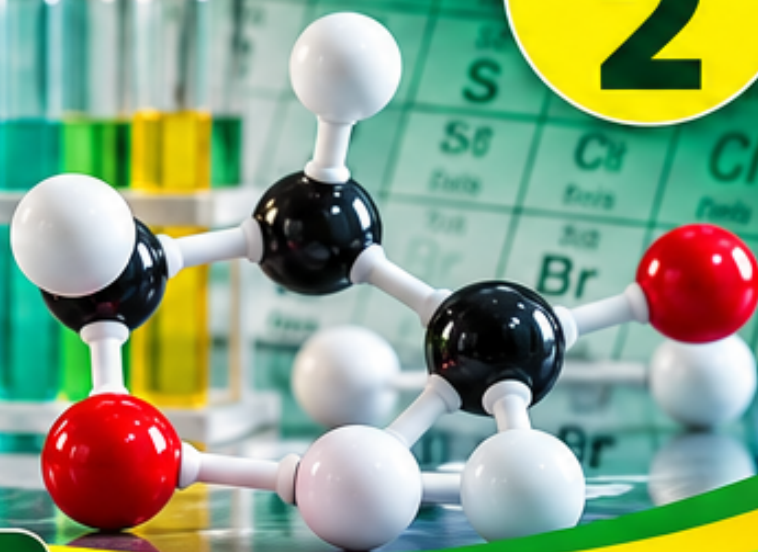


CHEMISTRY

**TOPICAL QUESTIONS &
NATIONAL EXAMINATION PRACTICE**

**BOOK
2**



THIS BOOK INCLUDES:



PART I – Topical Questions
(Chapters 1 – 7)



PART II – Topical Answers & Marking Guides
(Chapters 1 – 7)



PART III – National Examination Practice
(Papers 1 – 5, 80 marks each)



PART IV – Examination Marking Schemes
(Papers 1 – 5)



PART V – Final Revision Toolkit
(25 Revision Sections)

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SENIOR SCHOOL CHEMISTRY - GRADE 10

TOPICAL QUESTIONS & NATIONAL EXAMINATION PRACTICE

BOOK 2

Competency-Based Education (CBE)

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Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

— Hezekiah Omweri Nyatuka

How to Use This Book

REVISE → ATTEMPT → CHECK → CORRECT → RETRY

1. **REVISE** - Review the relevant concept in Book 1, class notes or your teacher's materials.
2. **ATTEMPT** - Answer the questions independently. Show working in calculations and use evidence in explanations.
3. **CHECK** - Compare your response with the marking guide only after completing the task.
4. **CORRECT** - Identify the exact scientific error, missing evidence, incorrect equation or weak reasoning.
5. **RETRY** - Attempt the question again without copying the answer.

Question Features

- **QUICK CHECK:** Short checks of essential understanding.
- **PRACTICE:** Structured questions for skill development.
- **APPLICATION:** Chemistry used in real-life or unfamiliar contexts.
- **PRACTICAL ACTIVITY:** Experimental planning, observations, variables, safety and evaluation.
- **HINT:** Strategic guidance without giving the full answer.
- **REMEMBER:** Essential facts or relationships worth retaining.
- **NOTE / NB:** Important distinctions and cautions.
- **EXAM FOCUS:** Precision expected in assessment responses.
- **THINK BIG:** Integrated higher-order reasoning and problem solving.

CHAPTER 1 - INTRODUCTION TO CHEMISTRY

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. Define Chemistry.
2. Name three branches of Chemistry.
3. A scientist determines the amount of iron in a water sample. Identify the branch of Chemistry involved.
4. State two applications of Chemistry in everyday life.
5. Name two careers that require Chemistry.
6. Explain one way Chemistry supports agriculture.
7. Explain one way Chemistry helps provide safe drinking water.
8. A farmer tests soil before applying fertiliser. State two benefits of the test.
9. A learner compares labels on two household cleaners. State two chemical ideas that can guide safe choice.
10. State two roles of Chemistry in the food industry.
11. Give one reason manufacturers test raw materials before production.
12. State one contribution of Chemistry to medicine.
13. State four laboratory safety rules.
14. You are given an unknown colourless liquid. State two safe observations/tests to begin investigating it.
15. A chemical container has no label. Explain what should be done.
16. A school produces acidic and alkaline waste solutions. Suggest a responsible approach before disposal.
17. Two chemicals are stored together although one label says 'oxidising' and the other 'flammable'. Explain the concern.
18. Explain what is meant by responsible use of Chemistry.
19. Plan a simple investigation of a community water-quality concern. State the question, one measurement and one safety rule.
20. Explain how Chemistry careers contribute to society.
21. THINK BIG: A community blames a factory for a change in river colour. State two kinds of evidence needed before reaching a conclusion.

CHAPTER 2 - THE ATOM

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. State the three main subatomic particles and their charges.
2. State where protons, neutrons and electrons are found in an atom.
3. Define atomic number.
4. Define mass number.
5. An atom has 11 protons and 12 neutrons. Determine its atomic number and mass number.
6. Complete the particle information for a neutral atom with atomic number 8 and mass number 16.
7. Define isotopes.
8. Explain why isotopes of an element have similar chemical properties.
9. Distinguish an atom from an ion.
10. An atom has 12 protons and 10 electrons. State its charge.
11. Write the shell electron arrangement of sodium, atomic number 11.
12. Write the shell electron arrangement of chlorine, atomic number 17.
13. State the maximum number of electrons in an s orbital.
14. State the maximum number of electrons in a p sublevel.
15. Write the s/p electron configuration of oxygen, $Z=8$.
16. Write the s/p electron configuration of magnesium, $Z=12$.
17. Write the s/p electron configuration of calcium, $Z=20$.
18. Explain why electrons fill lower-energy orbitals before higher-energy orbitals at this level.
19. An element has configuration $1s^2 2s^2 2p^6 3s^1$. Identify its number of occupied energy levels and outer electrons.
20. Calculate the relative atomic mass of isotopes 24 (68%) and 26 (32%).
21. Calculate the relative atomic mass of isotopes 63 (69%) and 65 (31%).
22. Calculate the relative atomic mass of isotopes 20 (90.5%) and 22 (9.5%).
23. An element has isotopes 10 and 11 and relative atomic mass 10.8. Explain what 10.8 means.
24. Particles L and M each contain 12 protons and 13 neutrons; L has 12 electrons while M has 10. Explain their relationship.
25. A learner says mass number and relative atomic mass are the same. Correct the statement.
26. State one piece of evidence that led scientists to revise atomic models.
27. THINK BIG: Explain why a scientific atomic model is useful even though an atom cannot be seen directly in an ordinary school laboratory.

CHAPTER 4 - CHEMICAL BONDING

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. Define ionic bonding.
2. Explain how a sodium ion forms.
3. Explain how a chloride ion forms.
4. Describe formation of sodium chloride in terms of electron transfer.
5. Define a covalent bond.
6. Explain formation of a single covalent bond in H_2 .
7. State the bonding in methane.
8. Define a dative covalent bond.
9. Use ammonium ion to illustrate dative bonding.
10. Define metallic bonding.
11. Explain why metals conduct electricity.
12. Explain why metals are malleable.
13. Explain why metals are ductile.
14. Describe the structure of a giant ionic lattice.
15. Explain the high melting point of sodium chloride.
16. Explain why solid sodium chloride does not conduct electricity.
17. Explain why molten sodium chloride conducts electricity.
18. Describe a simple molecular structure.
19. Explain why many simple molecular substances have low melting/boiling points.
20. State what is meant by intermolecular forces.
21. State what is meant by hydrogen bonding.
22. State what is meant by van der Waals forces at this level.
23. Explain why boiling water does not break O-H covalent bonds.
24. Describe diamond structure.
25. Explain why diamond is hard.
26. Explain why diamond does not conduct electricity.
27. Describe graphite structure.
28. Explain why graphite conducts electricity.
29. Explain why graphite is soft/slippery.
30. Compare diamond and graphite for one application each.
31. Describe the structure of a typical metal.
32. A substance has high mp, conducts as solid and molten, and is malleable. Identify its structure.
33. A substance has high mp, does not conduct as solid but conducts when molten. Identify its likely structure.
34. A substance has very high mp and does not conduct in either state. Suggest a likely structure.
35. Explain why high melting point alone cannot prove that a substance is ionic.
36. Compare ionic and covalent bond formation.
37. PRACTICAL/THINK BIG: State two measurable properties that can help distinguish ionic, metallic and simple molecular substances.
38. THINK BIG: A learner says 'all covalent substances are gases'. Evaluate.

CHAPTER 5 - PERIODICITY

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. Define periodicity.
2. State the general trend in atomic radius across a period.
3. Explain the atomic-radius trend across a period.
4. State the trend in atomic radius down a group.
5. Define first ionisation energy.
6. State the general first-ionisation-energy trend across a period.
7. Explain why first ionisation energy decreases down a group.
8. Define electron affinity.
9. Compare the size of a cation with its parent atom.
10. Explain why a cation is smaller than its parent atom.
11. Compare the size of an anion with its parent atom.
12. Explain why an anion is larger than its parent atom.
13. State the trend in Group 1 reactivity down the group.
14. Explain the Group 1 reactivity trend.
15. State the trend in Group 17 reactivity down the group.
16. Explain the Group 17 reactivity trend.
17. Arrange chlorine, bromine and iodine in decreasing reactivity.
18. State what happens when chlorine water is added to potassium bromide solution.
19. Write the ionic idea behind a halogen displacement reaction.
20. State the general trend in melting/boiling points as a periodic property.
21. State how electrical conductivity can vary across elements in a period.
22. Define ductility.
23. Define malleability.
24. Explain why periodic physical-property data should be interpreted alongside bonding and structure.
25. Describe a fair halogen-displacement investigation.
26. Atomic radii of successive elements are 186,160,143 pm. Describe the trend.
27. State two ways to improve reliability of periodic-property measurements.
28. Group 1 elements show increasing radius down the group. Explain how this supports increasing reactivity.
29. Group 17 elements show increasing radius down the group. Explain how this supports decreasing reactivity.
30. A learner says every periodic trend must be perfectly smooth. Evaluate.
31. Data show element A has a smaller radius than B in the same period. Give one possible explanation based on position.
32. An unknown halogen X displaces bromide ions but is not able to displace chloride ions. State its position in the reactivity sequence.
33. P, Q and R occur successively down Group 1. Their atomic radii increase $P < Q < R$. Which has the largest radius and why?
34. Explain why ionisation energy and atomic radius often show opposite general trends.
35. P, Q and R are in the same group with 2,3,4 occupied levels and one outer electron. State the radius, first-ionisation-energy and reactivity orders.

CHAPTER 6 - ACIDS AND BASES

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. Define an acid in aqueous solution.
2. Define a base.
3. Define an alkali.
4. State the relationship between bases and alkalis.
5. Define neutralisation.
6. Write the ionic equation for neutralisation.
7. State the pH range for acidic solutions.
8. State the pH of a neutral aqueous solution approximately.
9. State the pH range for alkaline solutions.
10. State the colour change of blue litmus in an acid.
11. State the colour change of red litmus in an alkali.
12. Explain the advantage of universal indicator over litmus for estimating acidity.
13. Write the equation for magnesium with hydrochloric acid.
14. State the test for hydrogen.
15. Write the equation for calcium carbonate with hydrochloric acid.
16. State the test for carbon dioxide.
17. Write the equation for copper(II) oxide with sulfuric acid.
18. Write the equation for hydrochloric acid with sodium hydroxide.
19. State the salt family formed from hydrochloric acid.
20. State the salt family formed from sulfuric acid.
21. State the salt family formed from nitric acid.
22. Explain why soil pH matters for crop growth.
23. State one way acidic soil may be treated.
24. Give one household application of acid-base chemistry.
25. Distinguish acid strength from acid concentration.
26. Explain why a strong acid of the same concentration may conduct better than a weak acid.
27. A weak acid is concentrated. Explain why 'weak' does not mean 'dilute'.
28. Plan an investigation comparing conductivity of equal-concentration strong and weak acids.
29. State the independent variable in an investigation of acid concentration and reaction rate with magnesium.
30. State two controlled variables in that investigation.
31. State one safety precaution when reacting acids with metals.
32. Explain why repeated measurements improve reliability.
33. A solution of pH 2 reacts with magnesium to produce hydrogen. What can be concluded safely?
34. State one additional test that could support identification of hydrochloric acid.
35. Explain why a neutralisation mixture is not automatically pH 7.
36. APPLICATION: A farmer finds soil pH 4.5 for a crop that grows best near pH 6.5. Suggest a chemical management step and one follow-up measurement.
37. THINK BIG: Why should pH, conductivity and reaction tests be combined rather than relying on one result to identify an unknown acid?

CHAPTER 7 - INTRODUCTION TO SALTS

Attempt all questions before consulting Part II. Use evidence, equations and clear scientific reasoning where required.

1. Define a salt.
2. Define a normal salt.
3. Define an acidic salt.
4. Define a basic salt.
5. Define a double salt.
6. State what is meant by solubility.
7. Distinguish soluble and insoluble salts.
8. State the preferred method for preparing an insoluble salt.
9. State the preferred method for a soluble salt from an acid and an insoluble base.
10. State the preferred method for a soluble salt from an acid and a soluble alkali.
11. Describe preparation of copper(II) sulfate crystals from copper(II) oxide and sulfuric acid.
12. Explain why copper(II) oxide is added in excess.
13. Explain why excess copper(II) oxide is filtered off.
14. Explain why the filtrate should not simply be evaporated to complete dryness.
15. Define crystallisation.
16. State the purpose of gentle heating during concentration.
17. State the purpose of cooling a concentrated solution.
18. State how crystals are collected.
19. State how crystals are dried.
20. Describe titration in salt preparation.
21. Explain why excess alkali cannot simply be filtered off in acid + soluble alkali preparation.
22. Define precipitation.
23. State the observation when $\text{Fe}^{2+}(\text{aq})$ is mixed with $\text{NaCl}(\text{aq})$.
24. Write the molecular equation for $\text{AgNO}_3 + \text{NaCl}$.
25. Write the ionic equation for silver chloride formation.
26. Define spectator ions.
27. Identify spectator ions in $\text{AgNO}_3 + \text{NaCl}$.
28. Write the ionic equation for barium sulfate formation.
29. Write the ionic equation for neutralisation.
30. Describe how an insoluble precipitate is purified after formation.
31. Explain why a precipitate is washed.
32. Explain why a precipitate is dried.
33. Define hygroscopic.
34. Define deliquescent.
35. Define efflorescent.
36. Distinguish hygroscopic and deliquescent behaviour.
37. Distinguish deliquescence and efflorescence.
38. State one useful application of salts in agriculture.
39. State one useful application of salts in medicine or daily life.
40. Explain why fertiliser salts should be applied in appropriate amounts.

PART II — TOPICAL ANSWERS & MARKING GUIDES

Use these answers to check concepts, scientific language and reasoning. Equivalent scientifically correct responses should be credited where appropriate.

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